

Online Python Compiler (Interp... xNew Tab x +

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Python Online Compiler

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main.py



Run

```
1 testcases=[[5,7,3,4,9,12,6,2],[1,3,5,7,9,11,13,15,17],[22,34,35,36,43,67,12,13,15,17]]
2 for case in testcases:
3     min_value=case[0]
4     max_value=case[0]
5     for i in case:
6         if(i<min_value):
7             min_value=i
8         if(i>max_value):
9             max_value=i
10    print("maximum value:",max_value,"minimum value:",min_value)
```

Output

Clear

maximum value: 12 minimum value: 2
maximum value: 17 minimum value: 1
maximum value: 67 minimum value: 12

=== Code Execution Successful ===

Activate Windows
Go to Settings to activate Windows.

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```
1 testcases=[[2,4,6,8,10,12,14,18],[11,13,15,17,19,21,23,35,37],[22,34,35,36,43,67,12,13,15,17]]
2- for case in testcases:
3     case.sort()
4     length=len(case)
5     min_value=case[0]
6     max_value=case[length-1]
7     print("minimum value:",min_value,"maximum value:",max_value)
```

Output

Clear

minimum value: 2 maximum value: 18
minimum value: 11 maximum value: 37
minimum value: 12 maximum value: 67

=== Code Execution Successful ===

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Run

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Clear

```
1 def mergesort(arr,lb,ub):
2     if(lb<ub):
3         mid=(lb+ub)//2
4         mergesort(arr,lb,mid)
5         mergesort(arr,mid+1,ub)
6         merge(arr,lb,mid,ub)
7 def merge(arr,lb,mid,ub):
8     temp=[0]*(ub-lb+1)
9     i=lb
10    j=mid+1
11    k=0
12    while(i<=mid and j<=ub):
13        if(arr[i]<arr[j]):
14            temp[k]=arr[i]
15            i=i+1
16        else:
17            temp[k]=arr[j]
18            j=j+1
19        k=k+1
20    while(i<=mid):
21        temp[k]=arr[i]
22        i=i+1
```

Output

[2, 4, 7, 9, 10, 12]

=== Code Execution Successful ===

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Run

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Clear

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1 def mergesort(arr,lb,ub):
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7 def merge(arr,lb,mid,ub):
8     temp=[0]*(ub-lb+1)
9     i=lb
10    j=mid+1
11    k=0
12    while(i<=mid and j<=ub):
13        if(arr[i]<arr[j]):
14            temp[k]=arr[i]
15            i=i+1
16        else:
17            temp[k]=arr[j]
18            j=j+1
19        k=k+1
20    while(i<=mid):
21        temp[k]=arr[i]
22        i=i+1
```

Output

[2, 4, 7, 9, 10, 12]

=== Code Execution Successful ===

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Run

Output

Clear

```
1 def partition(arr,lb,ub):
2     pivot=arr[lb]
3     start=lb
4     end=ub
5     while(start<end):
6         while(start<len(arr) and arr[start]<=pivot):
7             start=start+1
8         while(arr[end]>pivot):
9             end=end-1
10        if(start<end):
11            arr[start],arr[end]=arr[end],arr[start]
12    arr[lb],arr[end]=arr[end],arr[lb]
13    return end
14 def quicksort(arr,lb,ub):
15     if(lb<ub):
16         lock=partition(arr,lb,ub)
17         quicksort(arr,lb,lock-1)
18         quicksort(arr,lock+1,ub)
19 arr=[9,7,3,6,10]
20 n=len(arr)
21 quicksort(arr,0,n-1)
22 print(arr)
```

```
[3, 6, 7, 9, 10]

=== Code Execution Successful ===
```

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Run

Output

Clear

```
1 def partition(arr,lb,ub):
2     pivot=arr[lb]
3     start=lb
4     end=ub
5     while(start<end):
6         while(start<len(arr) and arr[start]<=pivot):
7             start=start+1
8         while(arr[end]>pivot):
9             end=end-1
10        if(start<end):
11            arr[start],arr[end]=arr[end],arr[start]
12    arr[lb],arr[end]=arr[end],arr[lb]
13    return end
14 def quicksort(arr,lb,ub):
15     if(lb<ub):
16         lock=partition(arr,lb,ub)
17         quicksort(arr,lb,lock-1)
18         quicksort(arr,lock+1,ub)
19 arr=[9,7,3,6,10]
20 n=len(arr)
21 quicksort(arr,0,n-1)
22 print(arr)
```

```
[3, 6, 7, 9, 10]

=== Code Execution Successful ===
```

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Run

```
1 def binarysearch(arr,n,key):
2     lb=0
3     ub=n-1
4     while(lb<=ub):
5         mid=(lb+ub)//2
6         if(key==arr[mid]):
7             return mid
8         if(key<arr[mid]):
9             ub=mid-1
10        else:
11            lb=mid+1
12    return -1
13 arr=[1,3,5,7,9,11]
14 key=7
15 result=binarysearch(arr,len(arr),key)
16 print(result)
17
18
```

Output

Clear

3

=== Code Execution Successful ===

JS

GO

php

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main.py

 Share  Run

```
1- def binarysearch(arr,n,key):
2-     lb=0
3-     ub=n-1
4-     while(lb<=ub):
5-         mid=(lb+ub)//2
6-         if(key==arr[mid]):
7-             return mid
8-         if(key<arr[mid]):
9-             ub=mid-1
10-        else:
11-            lb=mid+1
12-    return -1
13 arr=[1,3,5,7,9,11]
14 key=7
15 result=binarysearch(arr,len(arr).key)
16 print(result)
17
```

Output

Clear

3

=== Code Execution Successful ===

[illegible]

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Python

JS

GO

PHP

main.py

Run

Share

Output

Clear

```
1 A=[1,2]
2 B=[-2,-1]
3 C=[-1,2]
4 D=[0,2]
5 sum={}
6 for a in A:
7     for b in B:
8         s=a+b
9         if s in sum:
10             sum[s]+=1
11         else:
12             sum[s]=1
13 count=0
14 for c in C:
15     for d in D:
16         complement=-(c+d)
17         if complement in sum:
18             count+=sum[complement]
19 print(count)
```

2

=== Code Execution Successful ===